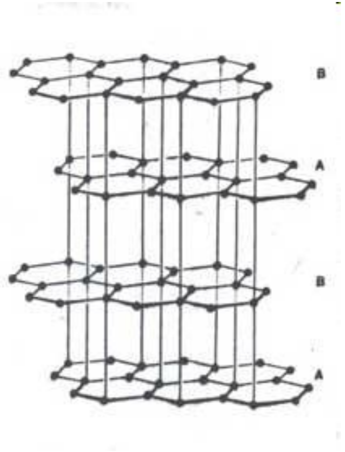
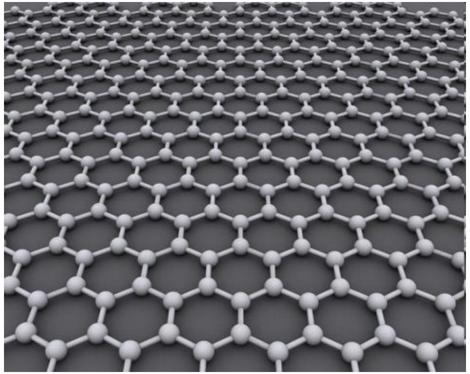
Modulation of analoques of unknown network

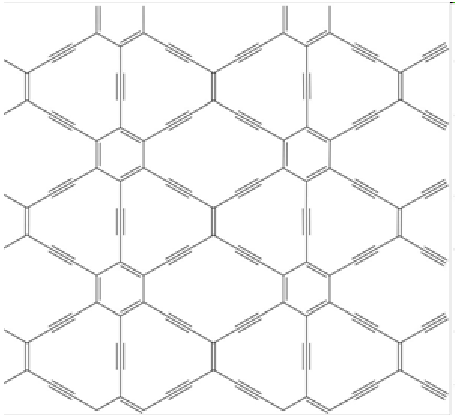
Graphite



Graphene



Graphyne



Graphene, a layer of graphite just one atom thick, isn't called a wonder material for nothing. The subject of the 2010 Nobel Prize in physics, it is famed for its superlative mechanical and electronic properties. Yet new computer simulations suggest that the electronic properties of a little-known sister material of graphene—graphyne—may in some ways be better.

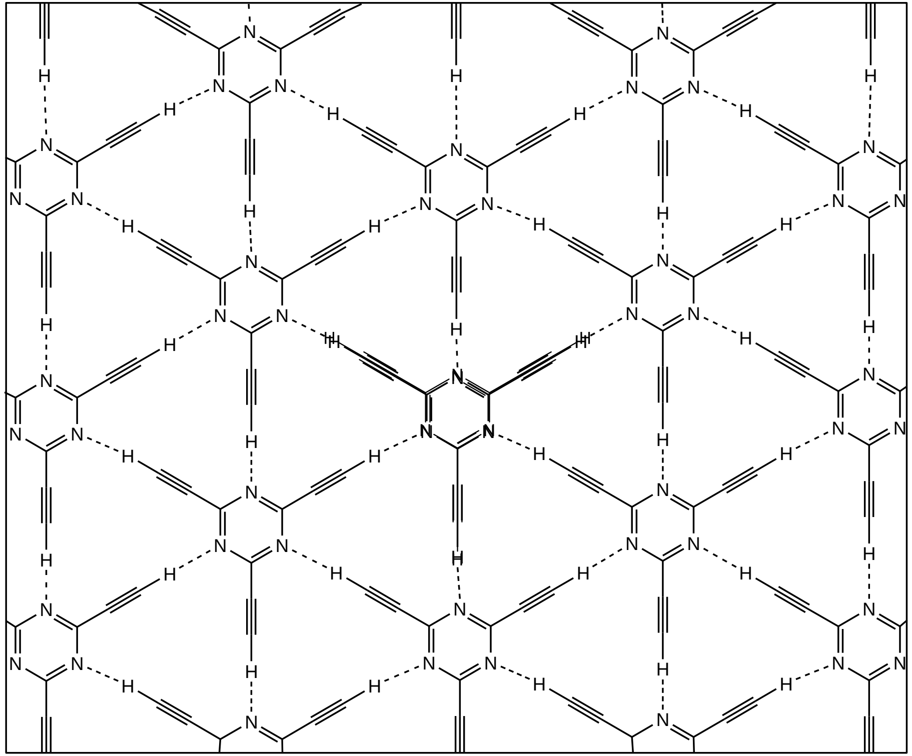
The simulations show that graphyne's conduction electrons should travel extremely fast—as they do in graphene—but in only one direction. That property could help researchers design faster transistors and other electronic components that process one-way current, says one of the study's authors, theoretical chemist Andreas Görling of the University of Erlangen-Nuremberg in Germany. "If your material already conducts in one direction, you have a head start," he says.

J. Cartwright, *Sciene Now*, **01.03.2012**

Supramolecular graphyne:

robust 2D hydrogen-bonded network structure with

suitable model: 2,4,6-triethynyl-1,3,5-triazine



- hexagonal array
- graphyne-like network
- 2D layers
- parallel stacking of layers
- weak C-H···N H-bonds
- <(C-H···N) = 180°
- all N and H atoms involved in the synthon formation

M. Ohkita, *Chem. Comm.* (**2002**) 3054

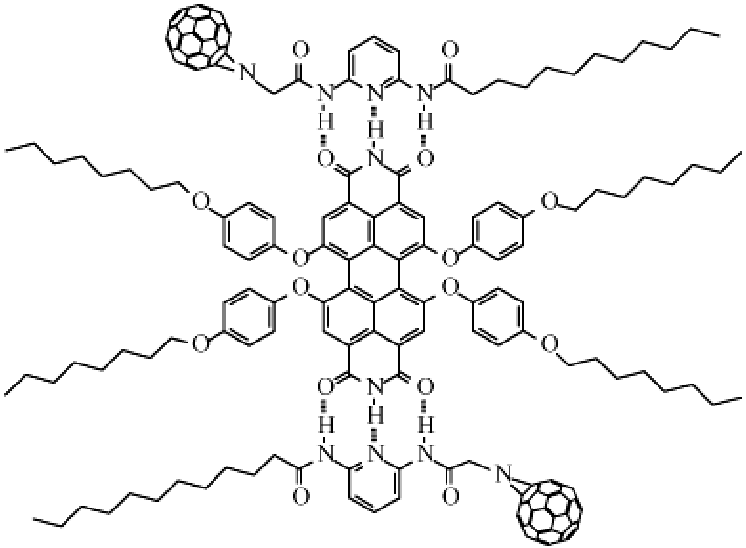
multiple hydrogen bonds for photovoltaic

Heinrich Heine
Universität Düsseldorf

Superstructure

Self-assembly of C_{60} fullerene derivative with PERY (perylene bisimide)

multiple HBs
DAD sequence



desired construction of functionalised fullerene-based architecture

Yang Liu, *J. Phys. Chem. B*, (**2004**), *108*, 6256.

Halogen Bonding (XB)…

SCIENTIFIC CONFERENCES

A big hello to halogen bonding

Halogen bonding connects a wide range of subjects — from materials science to structural biology, from computation to crystal engineering, and from synthesis to spectroscopy. The 1st International Symposium on Halogen Bonding explored the state of the art in this fast-growing field of research.

Mate Erdelyi

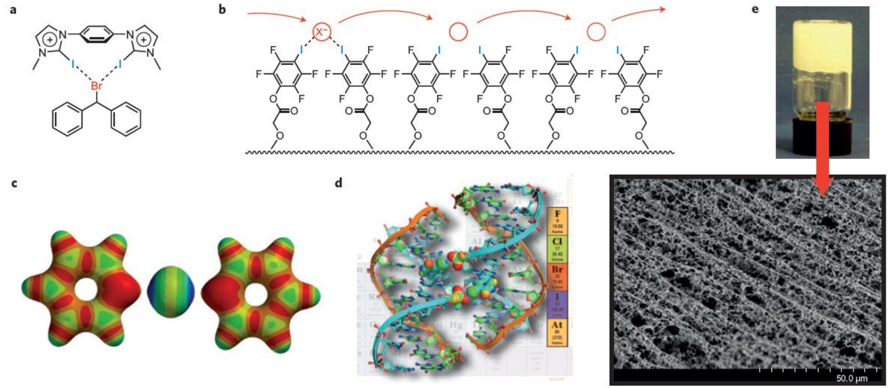


Figure 1 | Some highlights from the ISXB-1 meeting. **a**, A bidentate cationic halogen-bond donor designed for promoting halogen abstraction, for example in the solvolysis of benzhydril bromide. **b**, A synthetic ion channel that operates with halogen bonds. **c**, An iodonium ion is capable of forming two halogen bonds simultaneously, shown in the example of [bis(pyridine)iodine]$^+$. **d**, A DNA junction is an optimal model system to study biomolecular halogen bonds. **e**, An equimolar mixture of bis(pyridyl urea) and diiodotetrafluorobenzene in methanol–water forms a robust hydrogel upon rapid cooling. (Pui Shing Ho and Pierangelo Metrangolo are kindly acknowledged for providing figure parts **d** and **e**, respectively).

NATURE CHEMISTRY | VOL 6 | SEPTEMBER 2014 | www.nature.com/naturechemistry

IUPAC Definition

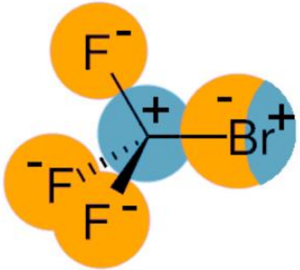
*“A **halogen bond** occurs when there is evidence of a net attractive interaction between an electrophilic region associated with a halogen atom in a molecular entity and a nucleophilic region in another, or the same, molecular entity”*

D-X···A **D-X···X-D**

(X = I, Br, Cl)
(A = O, N, S)

G. R. Desiraju, P. S. Ho, P. Metrangolo, G.Resnati, K. Rissanen et al . *Pure Appl. Chem.* **2013**, *85*, 1711.

Nature of halogen bond



XB arise from polarization of C-X bond

⇒ X is polarized positively in the region furthest away from the C atom

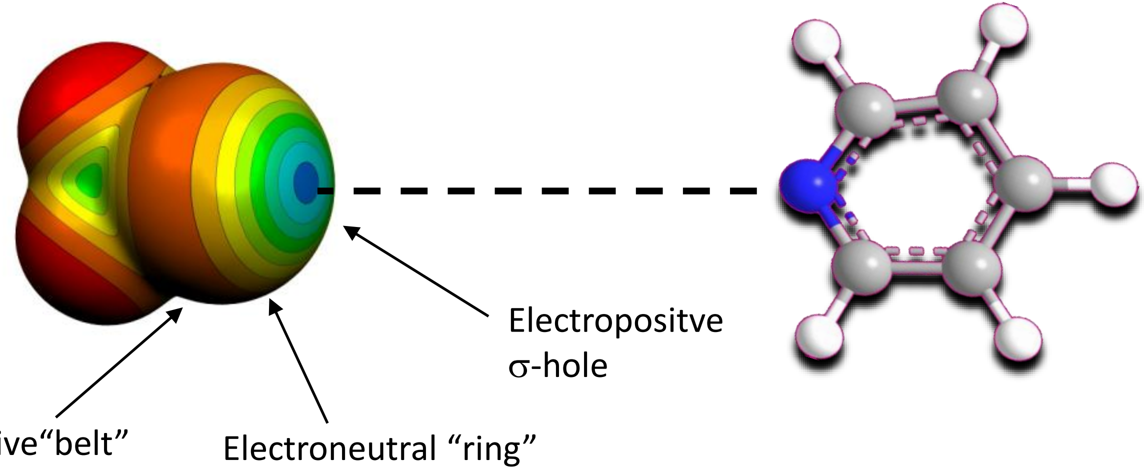
⇒ polarization gradient on C-X: $C^{\delta+}$- $X^{\delta-}$

⇒ Electrostatic contact with a negatively polarized species

⇒ Bond distance shorter than sum of VdW Radii

⇒ C-I···N angle approximately 180º

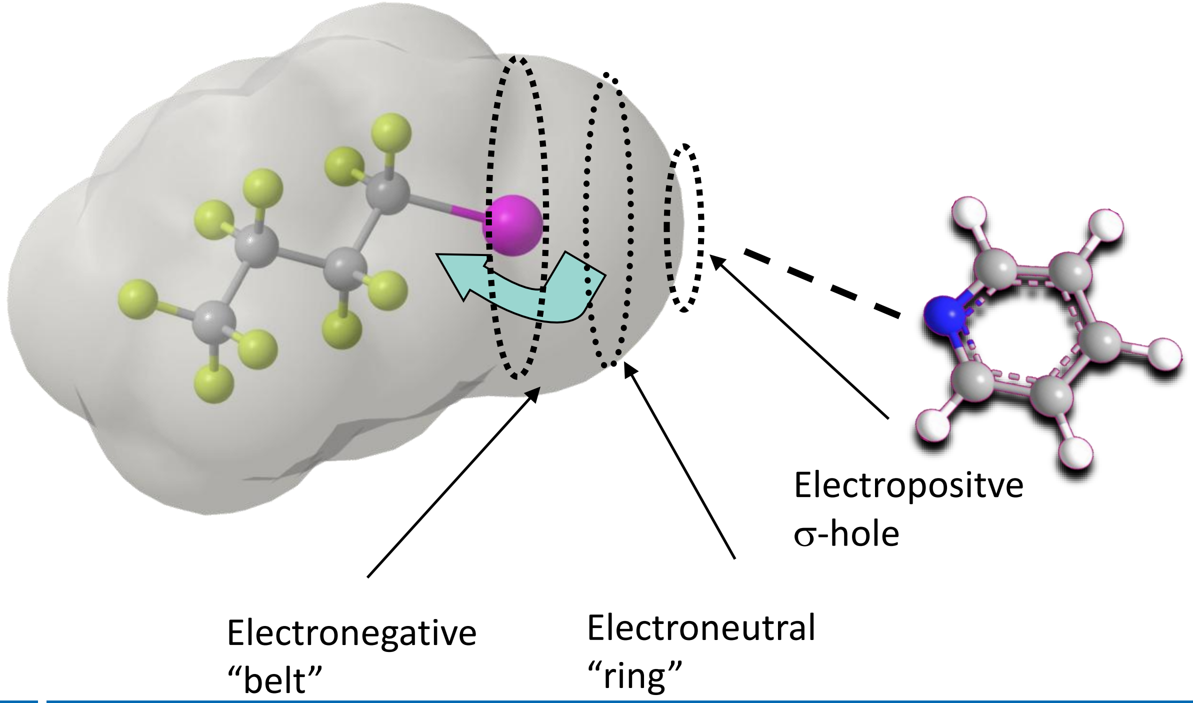
Electrostatic potential map
blue – positive region (**σ-hole**)
red – negative region



The σ-hole in halogen bonding

Heinrich Heine
Universität Düsseldorf

Perfluoroination: enlarges the σ-hole ⇒ interaction strength increases



- Polarization of C-X bond

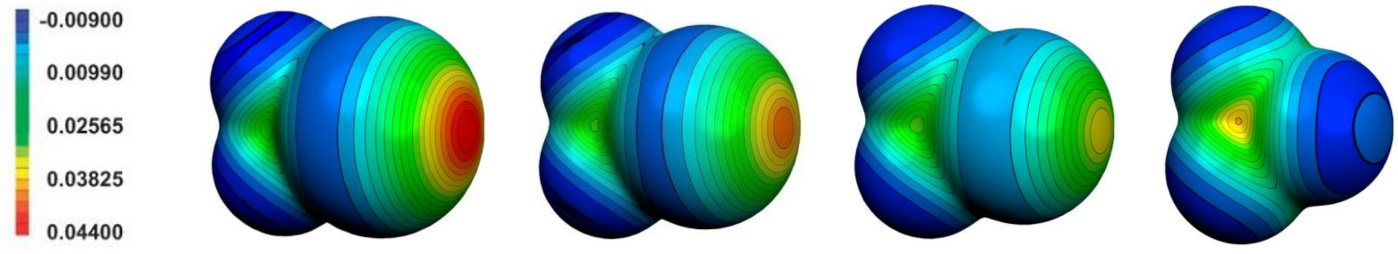
⇒ Electrostatic contact with a negatively polarized species

- Bond distance shorter than sum of VdW Radii
- **C-I···N** angle approximately 180°

The σ-hole dependence of halogen atom

Heinrich Heine Universität Düsseldorf

Perfluoroination: enlarges σ-hole ⇒ interaction strength increases



CF_3I CF_3Br CF_3Cl CF_4

Molecular electrostatic potential of CX_4 - halotrifluoromethane

Electrostatic potentials:
red – positive crown – **σ-hole**
blue – negative belt

positive region (σ-hole)

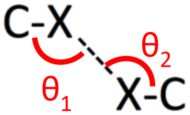
- intermolecular interaction strength I > Br >> Cl > F
- interacts with a N, O, Halogen atom:

C-I···N, C-I···O, C-Br···N, C-Br···N

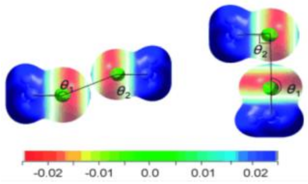
T. Clark, et al. *J. Mol. Model.*, **2007**, *13*, 291. P. Metrangolo, G. Resnati et. al. , *Angew. Chem. Int. Ed.*, **2008**, *47*, 6114

Halogen···halogen interaction and it‘s geometry

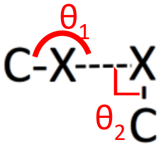
Calculated elektrostatic potentials
- blue – positive polarised,
- red – negatively polarised.



Type I



Type II



Geometry of two energy minimum of chloromethane

- symmetrical
- parallel, electrostatically identical regions
- van der Waals type
- shortest are actually repulsive
- $\theta_1 = \theta_2 \approx 150$
- Torsion angle (C–X···X–C), Φ = 180

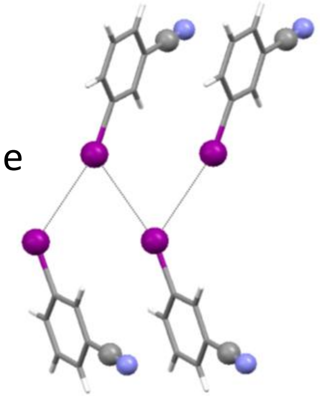
- genuine X-bond
- positive polarisation in the polar region of the halogen atom
- a negative polarisation in its equatorial region
- θ_1 = 180 °und θ_2 = 90 °

K. Merz, V. Vasylyeva, *CrystEngComm*, **2010**, *12*, 3989.

Properties of XB

Linear arrays in ***para***-iodobenzonitrile

meta-iodobenzonitrile
- zig-zag I···I chains



→ Molecular recognition dependant on substitution pattern

XB Properties

- directional -- σ-hole
- tunable interaction strength -- different depth of the σ-hole
- hydrophobic -- negative and positive sides
- large donor atom size -- tunable: I > Br > Cl

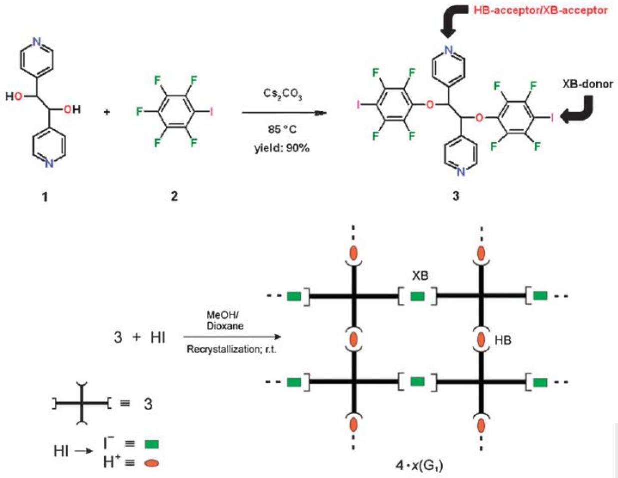
Merz, K., *Crystal Growth & Design*, **2006**, *6*, 1615.

P. Metrangolo, G.Resnati *Halogen Bonding: Applications and Fundamentals*, Springer Verlag, **2008**.

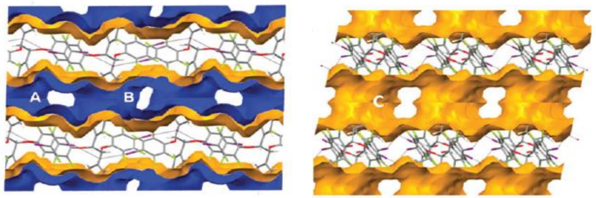
Hydrogen and halogen bonding drive the orthogonal self-assembly of an organic framework possessing 2D channels†

Heinrich Heine Universität Düsseldorf

- open framework with 2D channels
- able to host various guest molecules
- single-crystal-to-single-crystal guest exchange

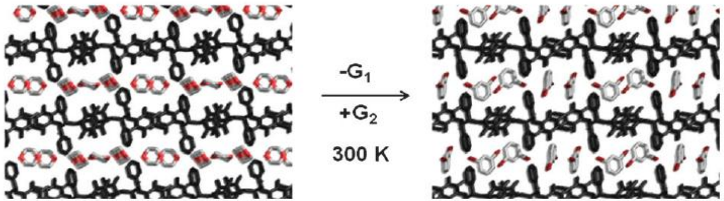


Voids observed after removal of guest molecules.



channels A and B

channel C intersects A and B channels



single-crystal-to-single-crystal guest exchange

XB a promising new tool for the design of open framework materials
→ pre-stage of non-covalently bonded metal-organic frameworks (MOF)

J. Marti-Rujas, L. Colombo, ..., G. Resnati, *Chem. Commun.*, **2012**, *48*, 8207. ; contact surface – yellow;,solvent accessible surface – blue

XB in luminescent materials

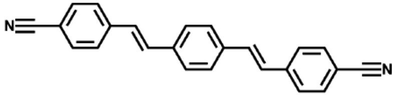
➤ self-assembling organic spin systems

➤ organic solid-state chromophores:

promising optoelectronic applications in the fields of light-emitting diodes, lasers, sensors, and two-photon fluorescent materials.

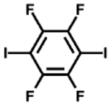
Challenge: obtain light-emission of an appropriate wavelength

fluorescent molecules **A**

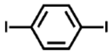


1,4-bis-*p*-cyanostyrylbenzene (A)

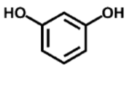
its co-formers **B-G**



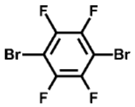
1,4-diiodotetrafluorobenzene (B)



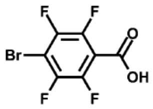
1,4-diiodobenzene (C)



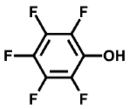
resorcinol (D)



1,4-dibromotetrafluorobenzene (E)



4-bromotetrafluorobenzene carboxylic acid (F)



2,3,4,5,6-pentafluorophenol (G)

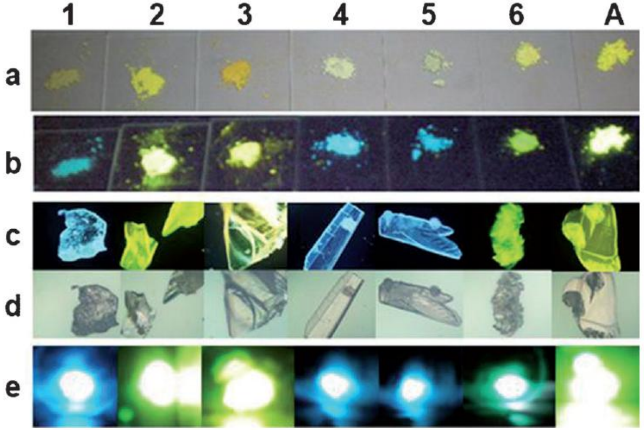


Figure 2. Photographs of solid-state cocrystal samples (from left to right: **1**–**6**) and **A**. a) and b) the powder samples under daylight and UV (365 nm); c) and d) the single crystal samples under UV (365 nm) and daylight observed by fluorescence microscope multiplied by 50-fold; e) two-photon luminescence under 800 nm laser.

Fine-tuning of UV/VIS absorption, luminescence emmision, color, lifetime, quantum yield and structural properties by cocrystal formation

D. Yan et. al, *Angew. Chem. Int. Ed.*, **2011**, *50*, 12483 –12486

XB in magnetic materials

➢ self-assembling organic spin systems

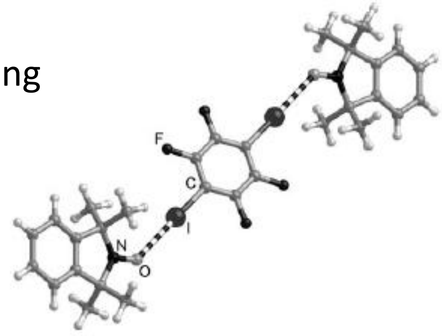
Heinrich Heine
Universität Düsseldorf

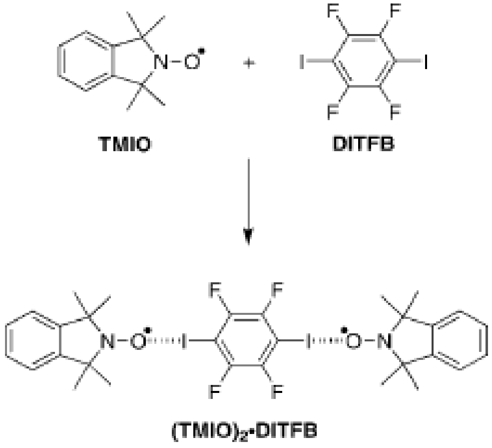
Iodoperfluorocarbons:

→ paramagnetic molecules as building blocks

Potential app.:

- molecular magnetism
- spintronics
- quantum computing





TMIO: tetramethylisoindolinyloxyl
DITFB: diiodotetrafluorobenzene

ionic resonance structure of the radical stabilized by N-O···I interaction

Maccaliff et. al, *Chem.Eur.J.*, **2009**, *15*, 4156

XB in fungicides

➢ case study of a preservative

3-Iodo-2-propynyl-*N*-butylcarbamate (**IPBC / Iodocarb**):

antimicrobial product used as preservative, fungicide and algaecide.

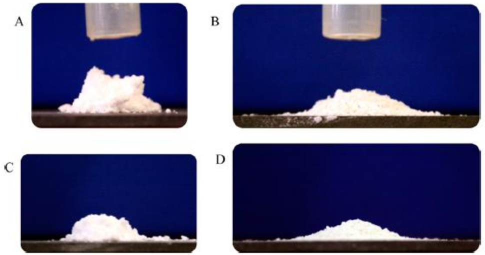


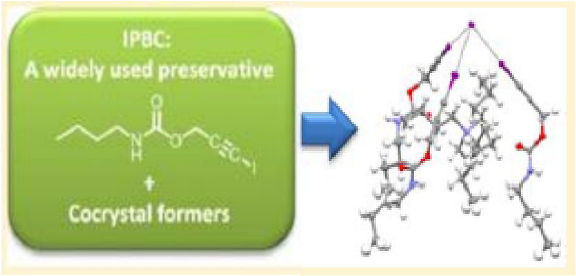
Figure 9. Pictures of cones of IPBC (A, C) and cocrystal 4 (B, D) powders, taken after flowing the powders through the funnel from 25 mm (A, B) and 50 mm (C, D) heights. The cylindrical shape of IPBC cones indicates clearly the high cohesion of the powders, while the flat cone shape of the cocrystal 4 indicates improved powder flow properties.

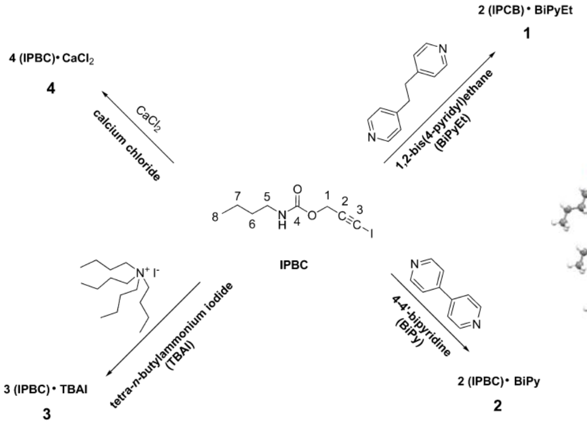
Problems:

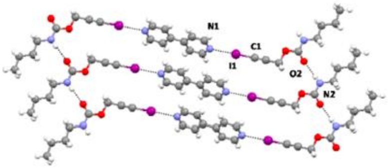
- Diffticult to obtain in pure form
- Hard to handle in industrial products and processes
- Sticky and clumpy

Idea:

- co-crystals







M. Baldrighi et al., *Mol. Pharmaceuticals*, **2013**, *10*, 1760

Does fluorine has an impact

1. on physico-chemical properties ?
2. on interactions ?

properties of F-atom:
- high electronegativity
- relatively small size
- very low polarisability

Perfluorocarbons

- Nearly ideal liquids.
- High compressibility.
- High vapor pressures.
- Poor solvents for organics.
  - non-polar molecules;
  - low polarizability;
  - small inter- and intra-molecular forces.

Hydrofluorocarbons

- Polar molecules.
- Appreciable inter- and intra-molecular interactions

F-substituent in **fluorocarbons**:
some tendencies

- Solubility (the least and the most polar solvents)
- Lipophilicity (increases in aromatic systems)
- Inductive effect (electron withdrawing)
- Acidity and basicity (increases HB acidity, decreases basicity)
- Steric effect (depends on compound and fluorination pattern)

Fluorine substituent effect

Heinrich Heine
Universität Düsseldorf

on physico-chemical properties of **fluorocarbons**:
only some tendencies

- Solubility (the least and the most polar solvents)
- Lipophilicity (increases in aromatic systems)
- Inductive effect (electron withdrawing)
- Acidity and basicity (increases HB acidity, decreases basicity)
- Steric effect (depends on compound and fluorination pattern)

Fluorine substituent effect

Table 1. Calculated hydrogen bond energies (kcal mol^{-1}) in some gas-phase dimers.[a]

Dimer	Energy
$[F{-}H{-}F]^-$	39
$[H_2O{-}H{-}OH_2]^+$	33
$[H_3N{-}H{-}NH_3]^+$	24
$[HO{-}H{-}OH]^-$	23
$NH_4^+ \cdots OH_2$	19
$NH_4^+ \cdots Bz$	17
$HOH \cdots Cl^-$	13.5
$O{=}C{-}OH \cdots O{=}C{-}OH$	7.4
$HOH \cdots OH_2$	4.7; 5.0
$N{\equiv}C{-}H \cdots OH_2$	3.8
$HOH \cdots Bz$	3.2
$F_3C{-}H \cdots OH_2$	3.1
$Me{-}OH \cdots Bz$	2.8
$F_2HC{-}H \cdots OH_2$	2.1; 2.5
$NH_3 \cdots Bz$	2.2
$HC{\equiv}CH \cdots OH_2$	2.2
$CH_4 \cdots Bz$	1.4
$FH_2C{-}H \cdots OH_2$	1.3
$HC{\equiv}CH \cdots C{\equiv}CH^-$	1.2
$HSH \cdots SH_2$	1.1
$H_2C{=}CH_2 \cdots OH_2$	1.0
$CH_4 \cdots OH_2$	0.3; 0.5; 0.6; 0.8
$C{=}CH_2 \cdots C{=}C$	0.5
$CH_4 \cdots F{-}CH_3$	0.2

On interactions?

Fluorine: no σ-hole → no XB formation

What about HB?

F atom should be a stronger H-bond-acceptor than O or N

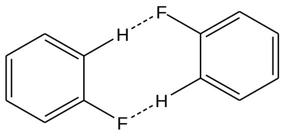
- correct for HF_2^-
- not correct for organic fluorinated compounds (C-F)

T. Steiner, *Angew. Chem. Int. Ed.* **2002**, *41*, 48

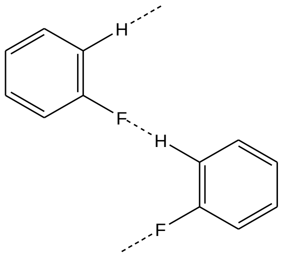
Fluorine substituent effect

On Interactions?

- C-H···F interactions similar to C-H···O und C-H···N, but very weak
- C-H···F well established but pretty unpredictable, system dependent



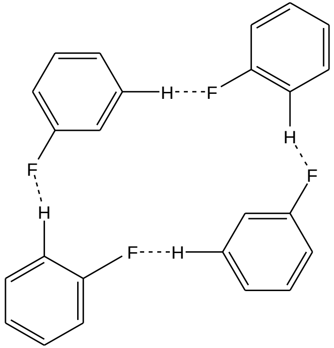
I



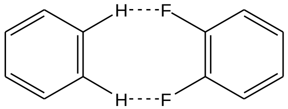
II

and still…

Classification of packing motifs with C-F···H interactions



III



IV

C-H···F comparable to C-H···N hydrogen bonds

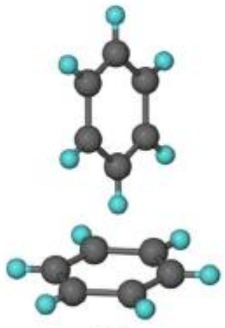
➢ stabilizing specific crystal structures

V. Vasylyeva, K. Merz *J. Fluorine Chem.*, **2010**, *131*, 446

π-interactions

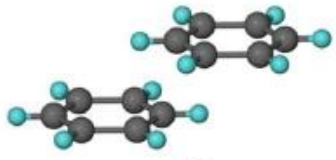
Geometrical classification of aromatic interactions

common also:
edge-to-edge
C-H···π
angle ≈ 45 - 60°
energy: < -8 kJ/mol



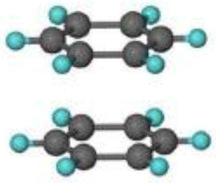
egde-to-face
T-shaped
C-H···π interaction

angle ≈ 90°
$d_{centroids}$: 3.6–4.0 Å
energy: < -12 kJ/mol
electrostatic contact
dispersion



parallel-displaced
slip-stacked, offset
π···π stacking

parallel ring planes
displacement ~1.5–2.0 Å
$d_{centroids}$: 3.3–3.8 Å
energy: up to -35 kJ/mol
electrostatic attraction
repulsion is reduced



sandwich geometry
face-to-face
π···π stacking

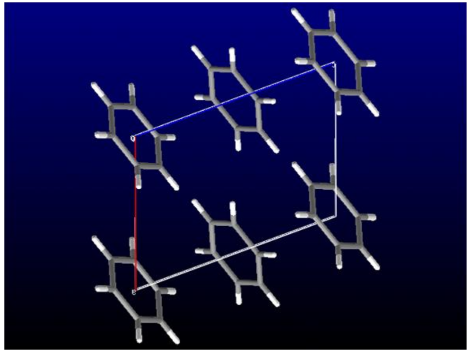
parallel ring planes
centroids aligned
$d_{centroids}$: 3.6–4.0 Å
energy: < -10 kJ/mol
high repulsion & dispersion

Podeszwa, R.; Bukowski, R.; Szalewicz, K., *J.Phys.Chem. A* **110,** 2006, 10345; M. O. Sinnokrot, C.D.Sherill, J. Phys. Chem. A 2004, 108, 10200

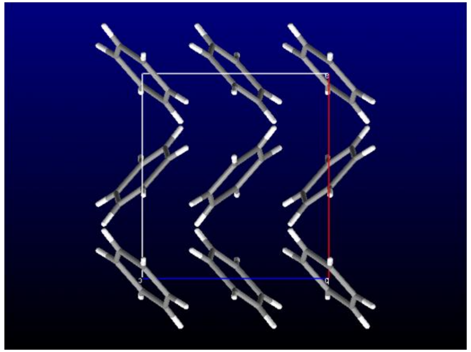
Crystal packing of benzene

➢ isostructural to fluorobenezene C_6F_6

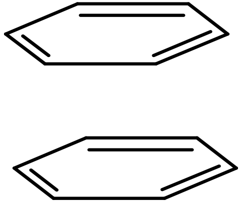
hhu
Heinrich Heine
Universität Düsseldorf



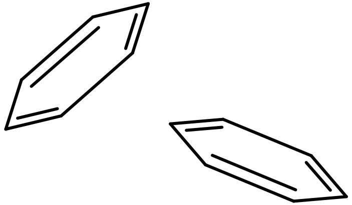
high pressure modification



modification at 1 atm



face-to-face π···π stacking



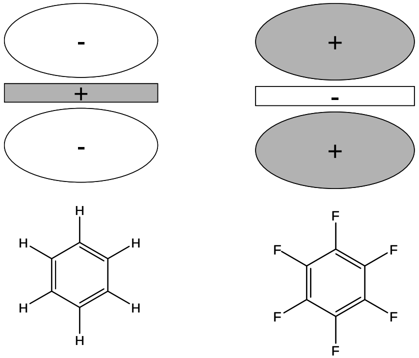
edge-to-face C-H···π interaction

Podeszwa, R.; Bukowski, R.; Szalewicz, K., *J.Phys.Chem. A* **110,** 2006, 10345; M. O. Sinnokrot, C.D.Sherill, J. Phys. Chem. A 2004, 108, 10200

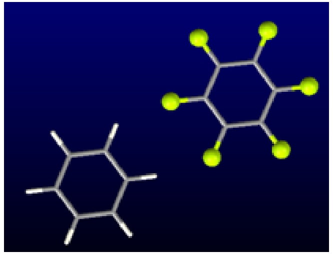
Crystal Engineering with C_6H_6 and C_6F_6

Quadrupol moments:

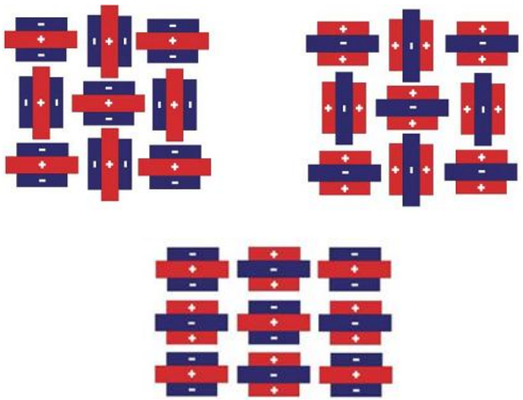
C_6H_6 C_6F_6



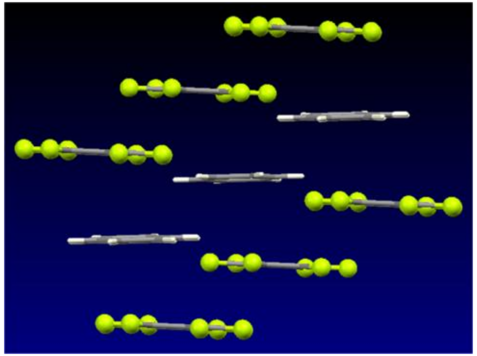
benzene hexafluorobenzene



C_6H_6 C_6F_6



C_6H_6 x C_6F_6 co-crystal



π···π stacking

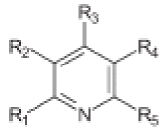
→ **aryl-perfluoroaryl interaction**

Hunter, C. A.; Sanders, J. K. M. *J. American Chemical Society* 1990, 112, 5525; Martinez, Iverson, *Chemical Science* **2012**, 3, 2191.

Aryl-perfluoroaryl interactions in CE co-crystal formation

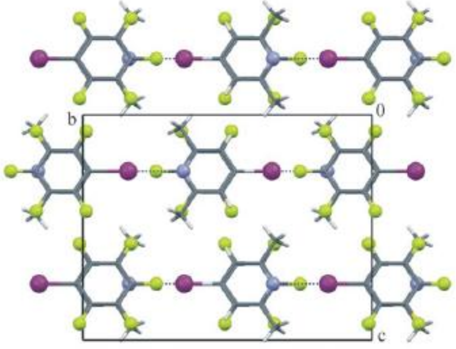
➢ XB + aryl-perfluorarryl stacking

C_6F_5I

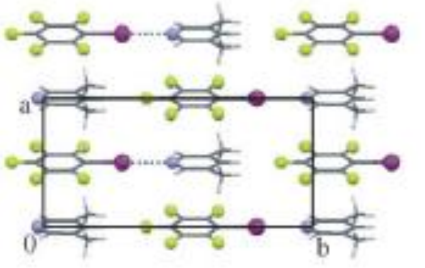


1

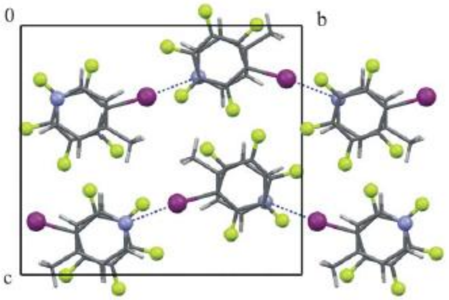
2 $R_1 = R_3 = R_5 = Me;\ R_2 = R_4 = H$
3 $R_1 = R_5 = Me;\ R_2 = R_3 = R_4 = H$
4 $R_2 = R_4 = Me;\ R_1 = R_3 = R_5 = H$
5 $R_3 = Me;\ R_1 = R_2 = R_4 = R_5 = H$



(a)



(c)

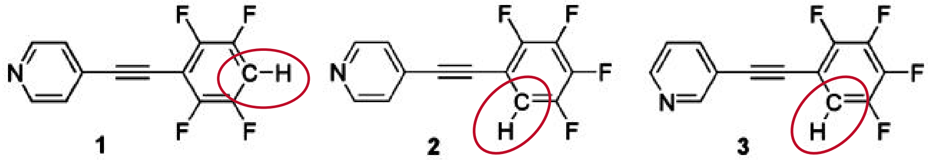


(b)

Polonski et. al, CrystEngComm, 2007, 9, 203–206

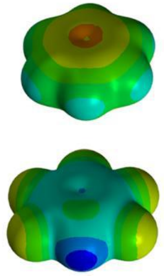
Aryl-perfluoroaryl interactions in CE

- HB + aryl-perfluoraryl stacking
- weak C-H···N HBs in F-substituted phenylethynylpyridines

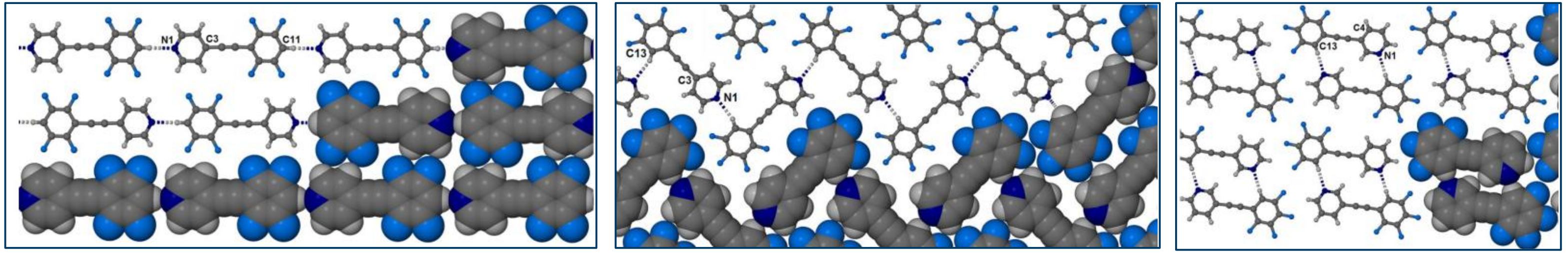


→ Increased acidity of phenyl proton by fluorination of the benzene ring

Electrostatic potential map of C_6H_6 and C_6HF_5



blue – positive charge



1: linear chains **2**: zig-zag chains **3**: dimers

E. Bosch, N. P. Bowling, J. Darko, *Crystal Growth & Design*, **2015**, *15*, 1634.

Fluorine applications in coatings and theranoctics

Heinrich Heine Universität Düsseldorf

Linear and branched perfluoropolyether coatings on diamond-like carbon films: covalent linkage and physical deposition

WALTER NAVARRINI[1], MAURIZIO SANSOTERA[1*], FRANCESCO VENTURINI[1], CLAUDIA L. BIANCHI[2], ANTONIO GUARDA[3], GIUSEPPE RESNATI[1]

Application of liquid lubricant films on diamond-like carbon (DLC) coatings.
Hard surfaces are typically lubricated with perfluoropolyethers (PFPE).

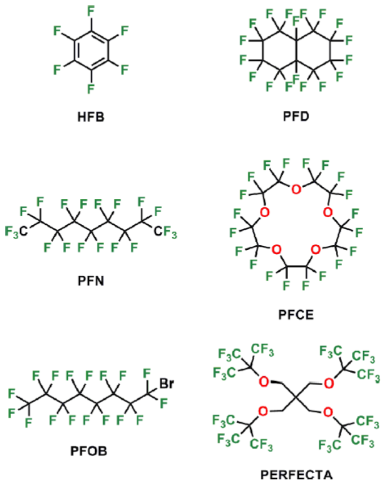
Water droplets on several DLC **coatings**

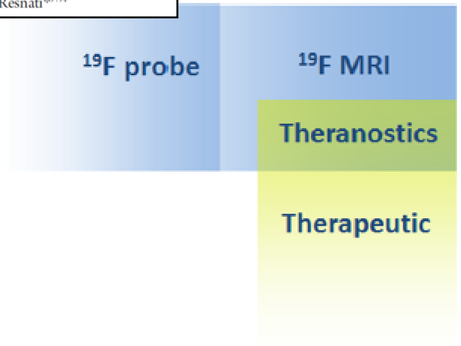
blank DLC thin film	linear PFPE coating	branched PFPE coating

CHEMICAL REVIEWS
Review
pubs.acs.org/CR

^{19}F Magnetic Resonance Imaging (MRI): From Design of Materials to Clinical Applications

Ilaria Tirotta,[†,‡] Valentina Dichiarante,[†,‡] Claudia Pigliacelli,[†,‡] Gabriella Cavallo,[†,‡] Giancarlo Terraneo,[†,‡] Francesca Baldelli Bombelli,[*,†,‡,§] Pierangelo Metrangolo,[*,†,‡,∥] and Giuseppe Resnati[*,†,‡]

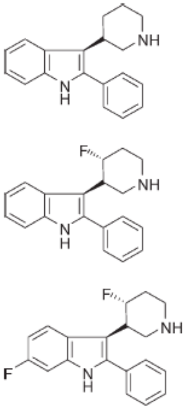




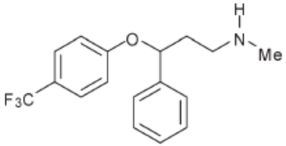
Polyfluorocarbons as **MRI tracers**: molecular tracers, polymers, branched derivatives

Fluorine in medicinal chemistry

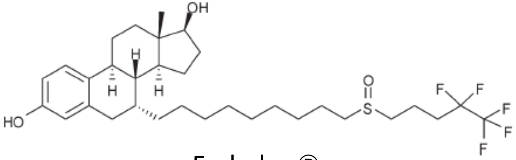
Heinrich Heine Universität Düsseldorf



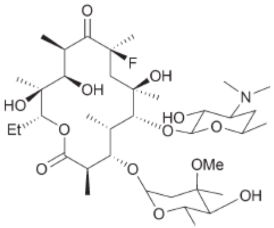
3-piperidinylindole
antipsychotic drugs



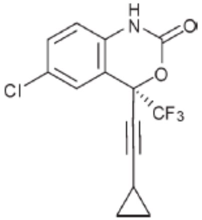
Prozac ®
antidepressant



Faslodex ®
oestrogen receptor antagonist



Flurithromicin
antibiotic



Efavirenz
transcriptase inhibitor HIV treatment

In a series of 3-piperidinylindole: fluorination decreased the basicity of the amine, thereby improving bioavailability

From Molecule to Material

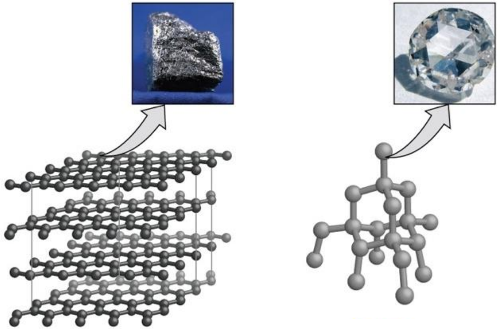
The controlled self-assembly of molecular species to give well defined supramolecular architectures is an important strategy for the development of functional molecular materials

Intermolecular interactions – useful tool for APPLICATIONS

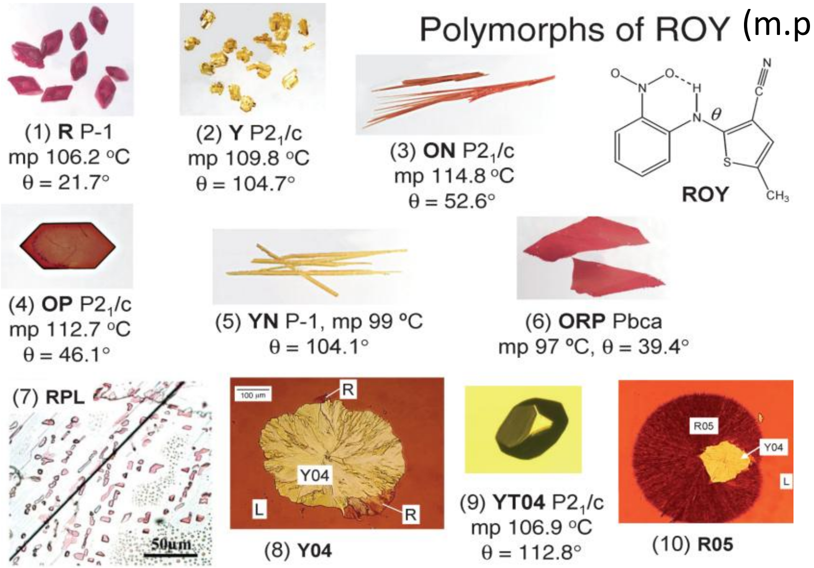
Polymorphism

„The Same and Not Quite the Same“ G. R. Desiraju

different crystal forms of the *same* chemical compound



(m.p.: 97-114. 8°)

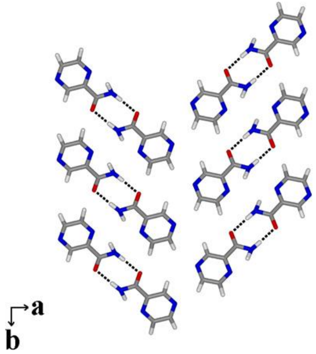


 G. R. Desiraju, *Cryst. Growth. Des.* **2008**, *8*, 3-5, J. Bernstein, *Cryst. Growth. Des.* **2011**, *11*, 632-650

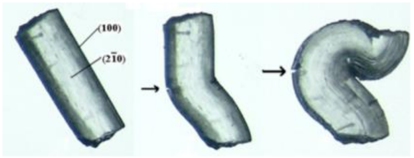
Properties of Polymorphs

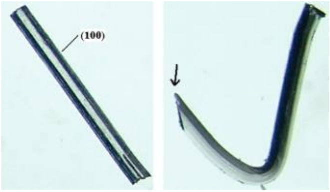
Heinrich Heine Universität Düsseldorf

- Mechanical Properties

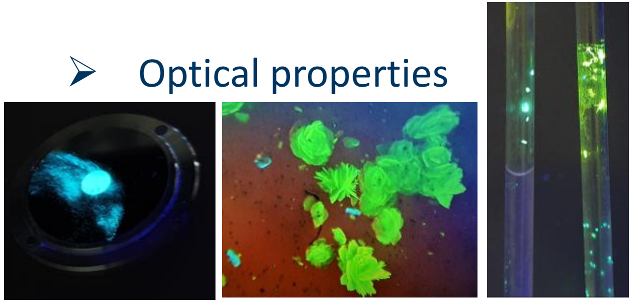


pyrazine-2-carboxamide

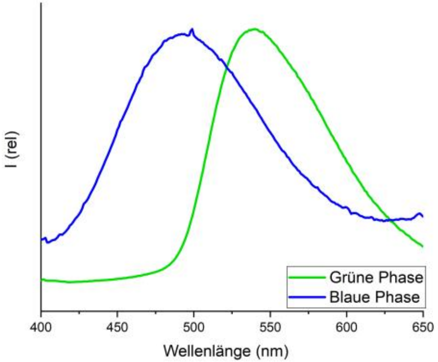




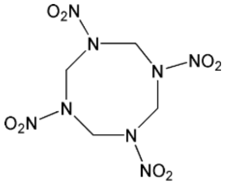
- Optical properties



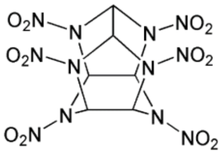
Form I **Form II**
of Cu(I)-HNC-complex
(from: bachelor thesis S. Hößel)



- Chemical Reactivity of Energetic Materials



HMX

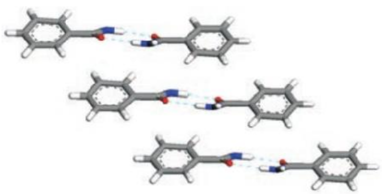


HNIW

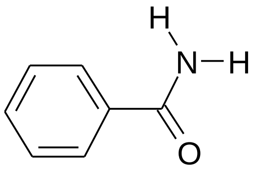
nitroamine explosive hexanitrohexaazaisowurtzitane

Polymorphism of benzamide – a historical case

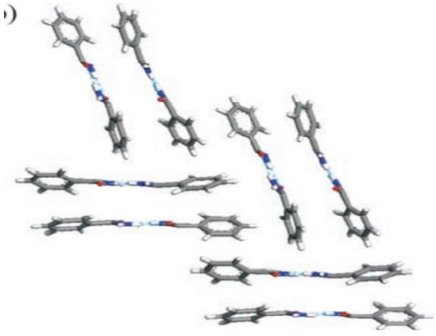
form I



$P2_1/c$ - X-Ray in 1959
- shifted π stacks

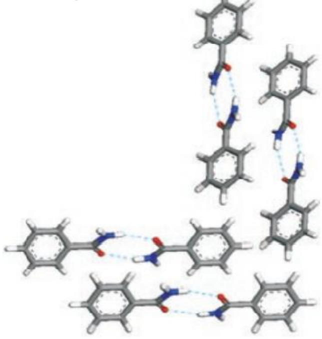


form II



*Pba*2 - X-Ray in 2005
- herringbone patterns
- pairs of H-bonded ribbons in T-shaped arrangement

form III



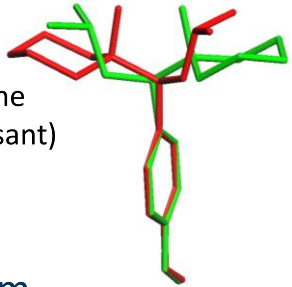
$P2_1/c$ - X-Ray in 2007
- herringbone patterns
- single ribbons T-shaped

The first incidence of polymorphism in molecular organic crystals based on their morphology.

F. Wöhler, J. Liebig, *Ann. Pharm*. **1832**, 249

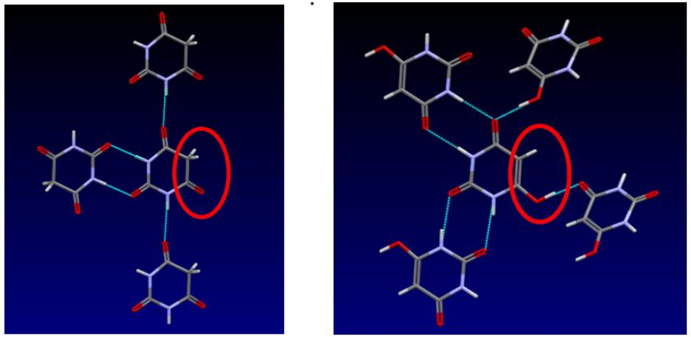
Types of Polymorphism

- **Conformation**al polymorphism



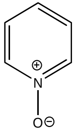
Venlafaxine (antidepressant)

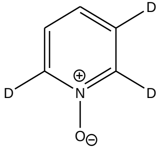
- **Tautomer**ic polymorphism

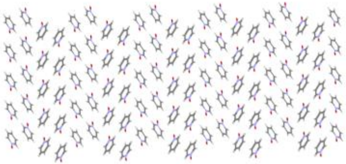


Keto-Enol-TP of barbituric acid

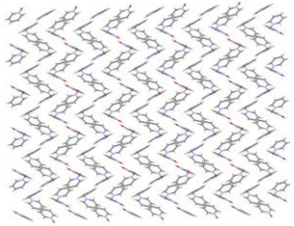
- **Isotop**omeric polymorphism







Form I

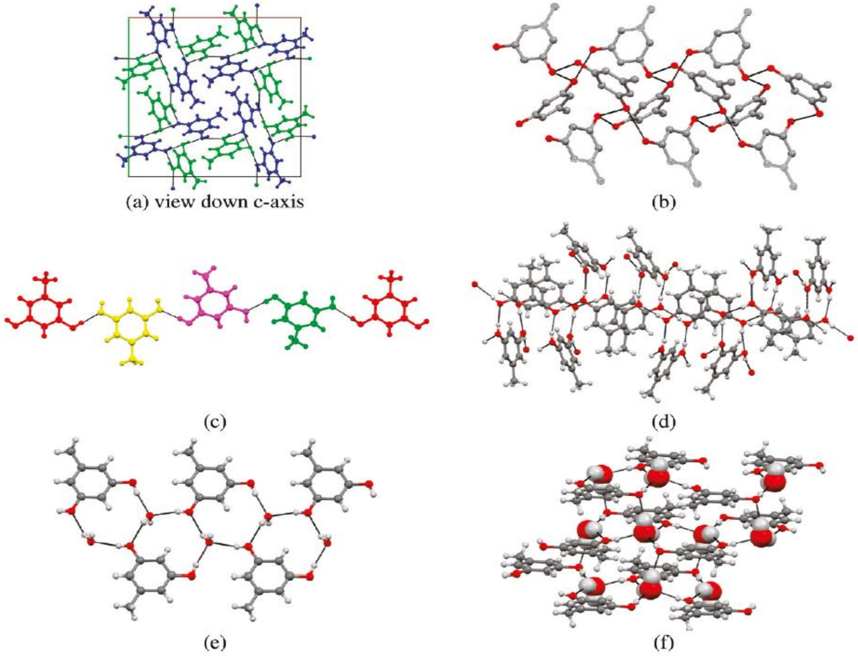


Form II

Changes in:
- overall packing , crystal system & space group
- intermolecular interactions

Pseudopolymorphism or Hydrates and Solvates

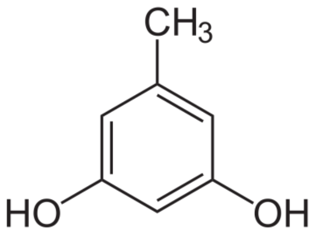
- organic molecule associated with different amounts of **solvent**
- multiple stoichiometries ($M \cdot H_2O$ / $M \cdot (H_2O)_{0.5}$ / $M \cdot (H_2O)_{1.5}$ / $M \cdot (H_2O)_2$) M = molecule)



form I

form II

monohydrate

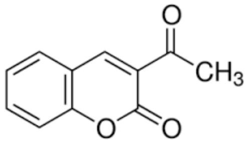


orcinol

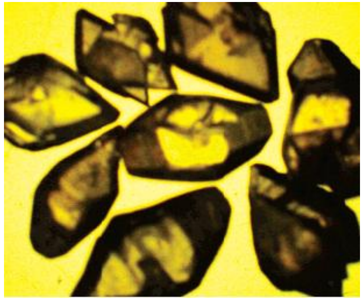
A. Mukherjee et al, *Cryst. Growth Des.*, **2011**, *11*, 2637-2653

Concomitant polymorphs

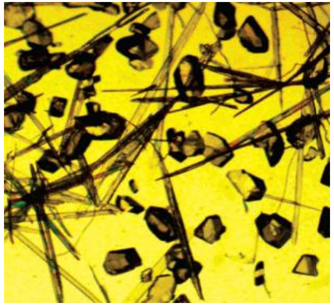
- Simultaneous appearance of different polymorphs in the same experiment under the same conditions



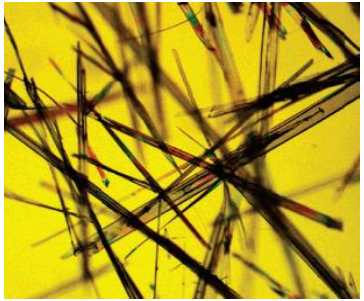
3-acetylcoumarin



Form A
23 °C
galcial acetic acid



Form A and B
5 °C
chloroform-*n*-hexane

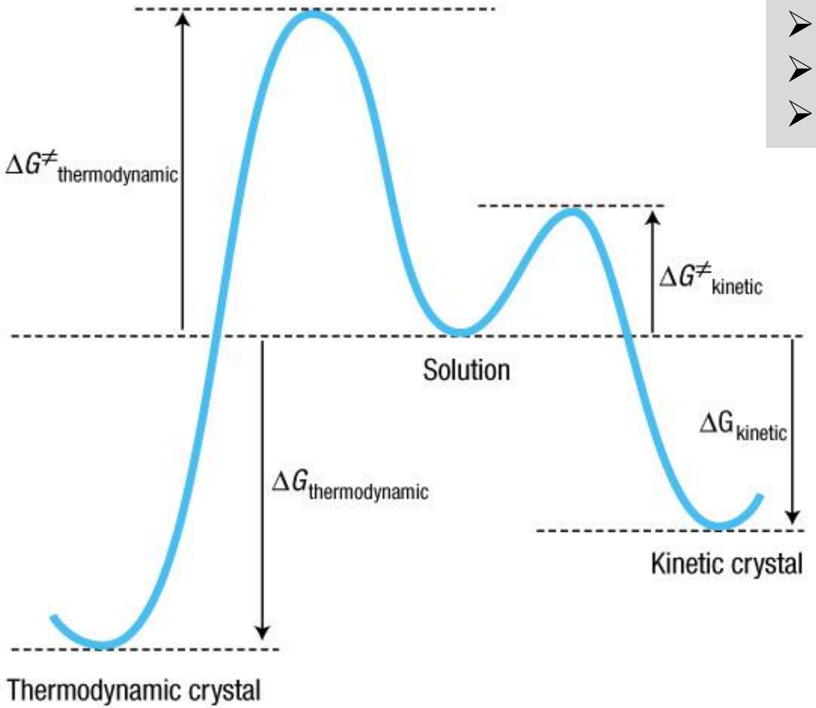


Form B
23 °C
chloroform-*n*-hexane

J. Bernstein, *Angew. Chem. Int. Ed.* **1999**, *38*, 3440-3461, T. N. Guru Row *et al.*, *Cryst. Growth. Des.* **2004**, *4*, 1105-1107

Thermodynamic and kinetic crystals

Heinrich Heine
Universität Düsseldorf



- Polymorphic systems: structures with similar lattice energies
- activation free energy are very similar
- appearance of the concomitant polymorphs possible

Ostwald's rule of stages

When leaving an unstable state, a system does not seek out the most stable state, but rather the nearest metastable state which can be reached with least loss of free energy

J. Bernstein, *Polymorphism in Molecular Crystals;* Oxford University Press: Oxford, **2002**

Factors for polymorph formation

Heinrich Heine Universität Düsseldorf

- Solvent
- Temperature
- Pressure
- Intermolecular interactions

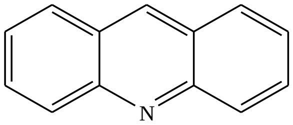
Result:

changes in the aggregation behaviour – **crystal packing**

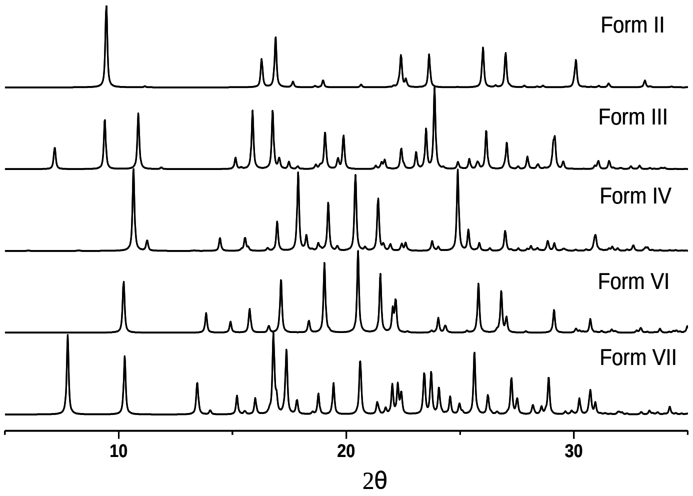
changes in intermolecular interactions – **supramolecular motives**

Solvent Effect

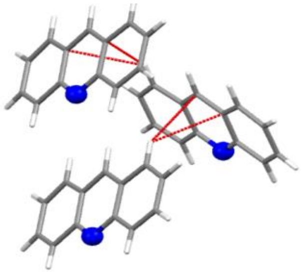
- on acridine



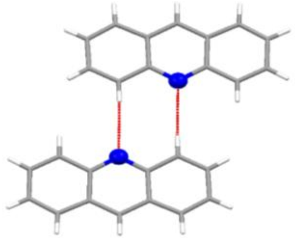
Check by PXRD



Check SC-XRD



Form II

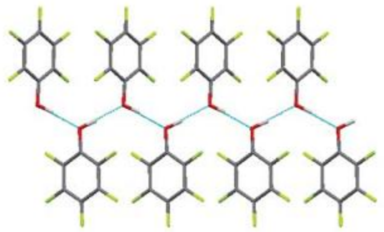


Form III

J McCabe et al.,*Cryst. Growth Des.*, **2010**, *10,* 1661-1664; X. Mei & C. Wolf, *Cryst. Growth Des.*, **2004**, *6,* 1099-1103; A. Kupka et al.,*Cryst. Growth Des*. **2012**, *12*, 5966–5971

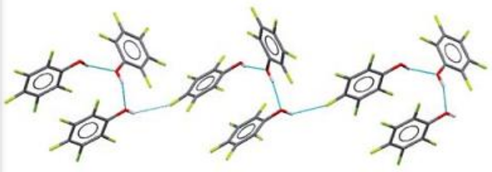
Temperature and pressure effect

pentafluorophenol



a

thermodynamic control
high temperature phase
T = 292(2) K



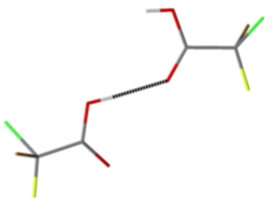
b

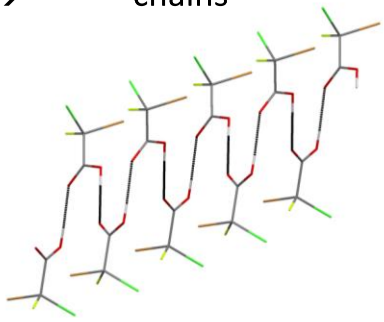
kinetic control
low temperature phase
T = 269(2) K

a) Form I with infinite ···OH···OH···OH··· chains
b) Form II with discrete OH···OH···OH trimer

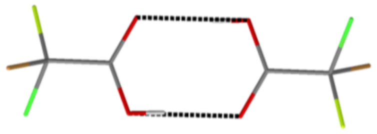
bromoclorofluoroacetic acid (CBrClFCOOH)

Phase α (0.59 GPa): catamer→ chains





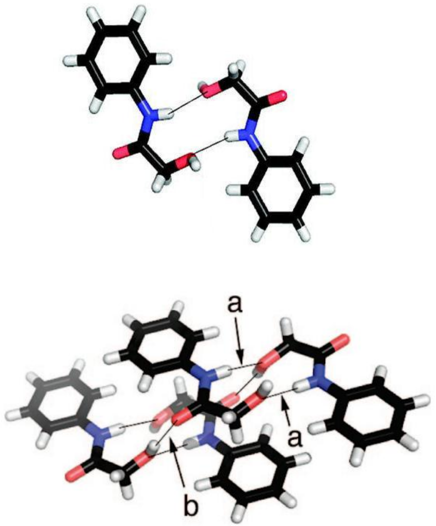
Phase β (1.37 GPa): dimer



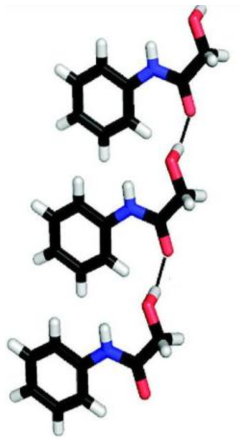
Under pressure molecules reorganise.
Same interactions result in different motives.

Intermolecular interactions in polymorphism

- HB dimer – catamer in glycolanilides



Metastable polymorph
- additional amide
  N−H bonded dimer



Thermodynamically stable polymorph
- O-H···O=C bonded catemer

General cases

- same functional group and the same motif but differences in the overall packing
- same functional group and the same motif but multiple occurrences of these groups in different molecular locations
- several functional groups and different motifs lead different packing arrangements

M. R. Hauser, *Crystal Growth & Design*, **2008**, *8*, 4428.

Application in Pharmaceutical Industry

Polymorphs display different solid-phase physical properties

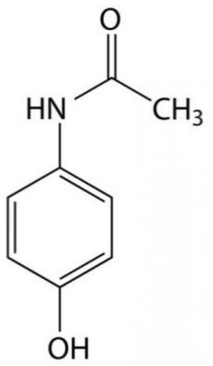
➢ polymorphism significantly affects properties of the solid form of a drug:

- solubility
- dissolution rate
- bioavailability/bioequivalence
- chemical stability …

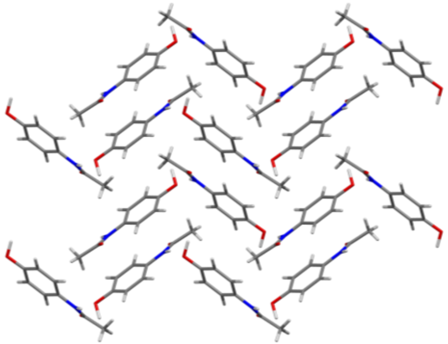
Polymorphism of Acetaminophen

Paracetamol

- ➢ an analgesic drug



Form I ($P2_1/a$)

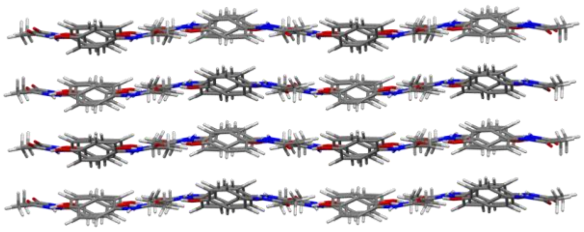


- thermodynamic polymorph
- commercially sold
- herringbone arrangement
- tableting problems
- no direct compression

→ binding agents

➢ costly in time and materials

Form II (Pcab)



- kinetic, metastable polymorph
- well-developed slip planes
- plastic deformation upon compaction
- direct tableting
- slightly more soluble than form I

➢ processing advantages & commercial benefit

Form II is better! Why is it still not on the market?

 X. Wang & D. J. Kirwan, *Cryst. Growth Des.*, **2006**, *6*, 2228-2240

Disapearing polymorphs: Ranitidine·HCl block buster

- H2-antihistaminikum
- anti-ulcer drug
- prevent and relieve heartburn

1977: synthesis, 1st rational drug design

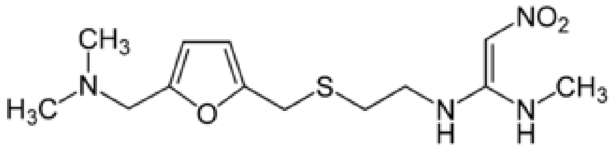
1978: “plate patent” of **form I** (**valid till 1995**)

- **Zantac**: introduced, licensed and marketed

Some time later…

in the 13th production charge:

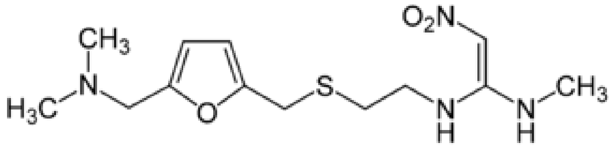
→ new „needle“ **form II**



Pure phase I is not producable at the Glaxo factory since then

1985: new “needle patent” of form II

- marketing



Ranitidine·HCl

a patent protection and extention block buster

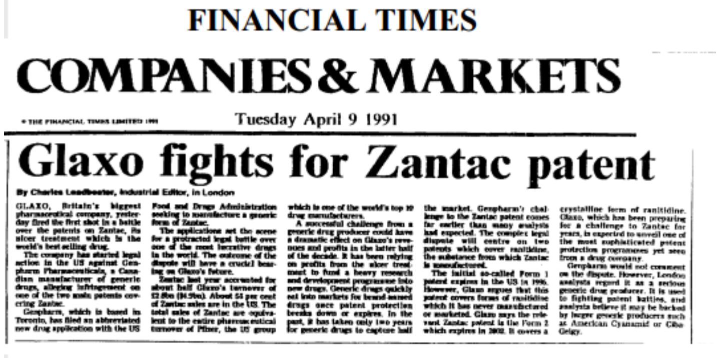
Early 90-s: the sales reached 3.8 billion \$/year = 50 % of whole merit of Glaxo group

New player:

a generic pharma company **Novopharm**:

- could not reproduce the patented form I
- claim „needle patent“ to be invalid

➢ Novopharm could market form II



FINANCIAL TIMES

COMPANIES & MARKETS

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Glaxo fights for Zantac patent

By Charles Leadbeater, Industrial Editor, in London

GLAXO, Britain's biggest pharmaceutical company, yesterday fired the first shot in a battle over the patents on Zantac, its ulcer treatment which is the world's best-selling drug.

The company has started legal action in the US against Genpharm Pharmaceuticals, a Canadian manufacturer of generic drugs, alleging infringement on one of the two main patents covering Zantac.

Genpharm, which is based in Toronto, has filed an abbreviated new drug application with the US Food and Drugs Administration seeking to manufacture a generic form of Zantac.

The applications set the scene for a protracted legal battle over one of the most lucrative drugs in the world. The outcome of the dispute will have a crucial bearing on Glaxo's future.

Zantac last year accounted for about half Glaxo's turnover of £2.8bn (\$4.9bn). About 54 per cent of Zantac sales are in the US. The total sales of Zantac are equivalent to the entire pharmaceutical turnover of Pfizer, the US group which is one of the world's top 10 drug manufacturers.

A successful challenge from a generic drug producer could have a dramatic effect on Glaxo's revenues and profits in the latter half of the decade. It has been relying on profits from the ulcer treatment to fund a heavy research and development programme into new drugs. Generic drugs quickly eat into markets for brand-named drugs once patent protection breaks down or expires. In the past, it has taken only two years for generic drugs to capture half the market. Genpharm's challenge to the Zantac patent comes far earlier than many analysts had expected. The complex legal dispute will centre on two patents which cover ranitidine, the substance from which Zantac is manufactured.

The initial so-called Form 1 patent expires in the US in 1996. However, Glaxo argues that this patent covers forms of ranitidine which it has never manufactured or marketed. Glaxo says the relevant Zantac patent is the Form 2 which expires in 2002. It covers a crystalline form of ranitidine. Glaxo, which has been preparing for a challenge to Zantac for years, is expected to unveil one of the most sophisticated patent protection programmes yet seen from a drug company.

Genpharm would not comment on the dispute. However, London analysts regard it as a serious generic drug producer. It is used to fighting patent battles, and analysts believe it may be backed by larger generic producers such as American Cyanamid or Ciba-Geigy.

Glaxo sue Novopharm and defends:

- an **independent lab** can **reproduce** the protected procedure of **phase I** formation

Novopharm lost the first battle!

Ranitidine·HCl

a patent protection and extention block buster

The show goes on…

1994: **Novopharm** – new preparation method for the vanished pure form I

➢ wanted: new pantent and marketing

Glaxo sue Novopharm again and claim:

- form I is not pure

- contains form II (protected by the „needle patent")

Novopharm proves the purity of form I

➢ market a genericum Novo-Ratinidine

First major legal event!

Brought scientific field of organic polymorphism into the glare of commercial and industrial activities

Cocoa Butter Polymorphism

Heinrich Heine Universität Düsseldorf

Six polymorphs:

I	rapid cooling of the melt	m.p. 17.3 °C
II	rapid cooling of the melt with 2 °C/min	m.p. 23.3 °C
III	crystallization at 5-10 °C	m.p. 25.5 °C
IV	crystallization at 16-21 °C	m.p. 27.3 °C
V	slow crystallization	m.p. 33.8 °C
	⇒ ***desired edible form***	
VI	from form V after several months at RT	m.p. 36.3 °C
	⇒ ***thermodynamic stable form; grainy, dull surface, does not melt in the mouth***